

ISHAN JAIN

+91 9039702820 | i8han236@email.com | GitHub | LeetCode

Education

Birla Institute of Technology, Mesra

B.Tech in Computer Science & Engineering

2024 – 2028

CGPA: 8.51/10

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

Projects

Relay – Distributed Task Queue

Node.js, TypeScript, Express, PostgreSQL, Redis, WebSockets, Next.js, Docker

- Designed and implemented a distributed task queue supporting reliable background job execution using PostgreSQL, Redis Pub/Sub and WebSockets.
- Implemented safe concurrent job claiming using `SELECT ... FOR UPDATE SKIP LOCKED` inside transactions, enabling multiple workers to process jobs without duplication.
- Built a fault-tolerant scheduling pipeline featuring priority queues, configurable retry policies, exponential backoff (30s → 600s) and automatic dead-letter queue routing.
- Implemented worker recovery logic to reclaim jobs abandoned by crashed workers, preventing indefinite RUNNING states and improving queue reliability.
- Developed a real-time monitoring dashboard using Next.js that streams live job state changes through Redis Pub/Sub and WebSockets.

Multi-Tenant E-Commerce Platform

Next.js, TypeScript, PostgreSQL, Prisma, NextAuth.js, Razorpay, Cloudinary

- Built a multi-tenant e-commerce platform enabling two independent storefronts to operate from a shared Next.js application while maintaining isolated inventory, orders and storefront configuration.
- Designed a normalized PostgreSQL schema for products, variants, add-ons and orders, modelling order lifecycle using state-machine based status transitions.
- Implemented secure authentication using NextAuth.js with Google OAuth, JWT-based sessions and role-based access control enforced through middleware.
- Developed an admin dashboard supporting product CRUD, inventory management and Cloudinary-powered image uploads with primary image selection.

Skills

Languages: C++, C, TypeScript, JavaScript, SQL

Frameworks: React, Next.js, Express

Backend: REST APIs, WebSockets

Databases: PostgreSQL, Redis, Prisma ORM

Developer Tools: Git, Docker, Postman, Vercel

Coding Achievements

Solved **350+** algorithmic problems across LeetCode and Codeforces covering graphs, dynamic programming, greedy algorithms and advanced data structures.

LeetCode Rating: **1630+**

Codeforces Rating: **1100+**